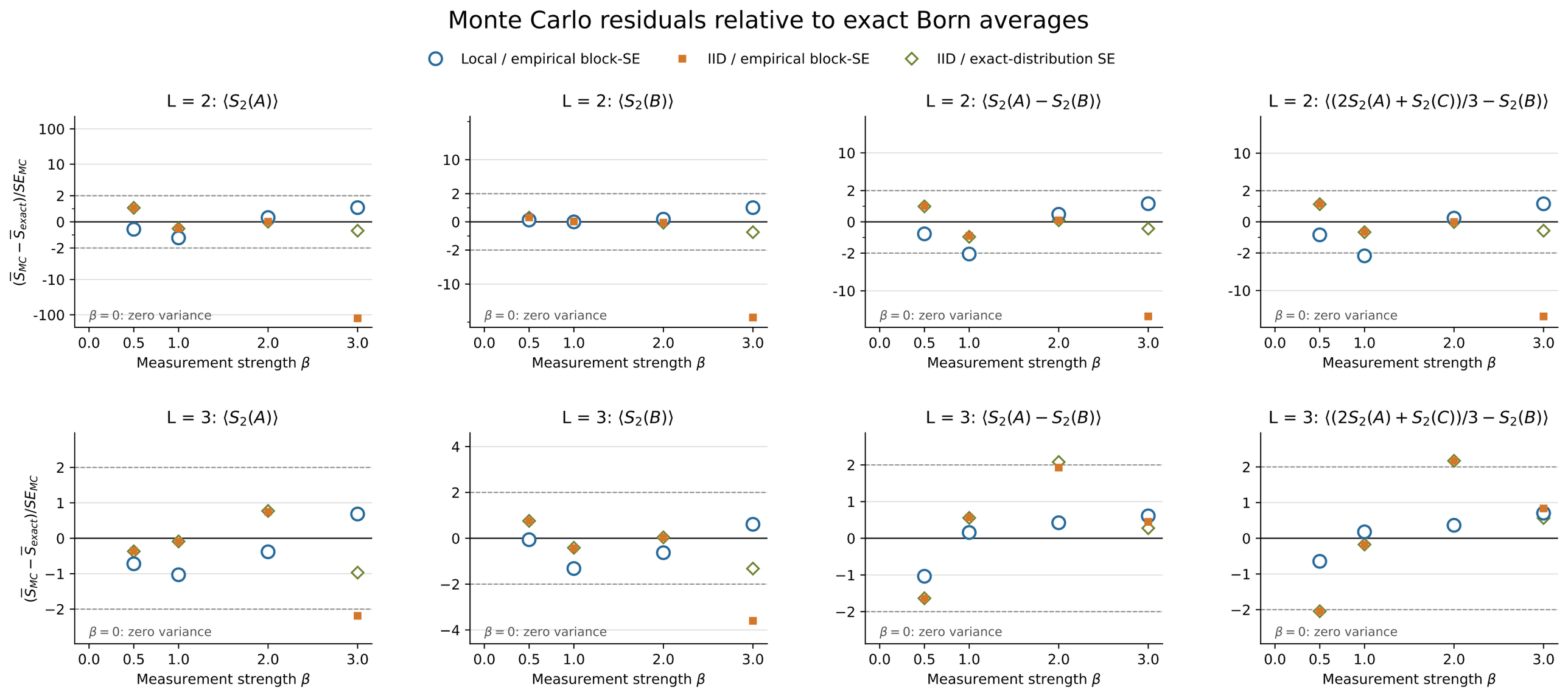

L2 y-axis is symmetric-log (linear within |z| <= 3), preserving rare-tail outliers. L3 y-axis is linear; +/-2 lines are reference levels.
Exact-distribution SE = sqrt(Var_exact / 30,000) for IID only; a finite sample can miss rare outcomes and underestimate its empirical SE.
At beta=0 variance is zero: z is undefined and omitted. These multiple residuals are diagnostics, not a simultaneous confidence test.